MANAV SHAH

AI Engineer | Generative AI | LLMs | Machine Learning

Manavshah052003@gmail.com | 8128362204 | www.github.com/manavshah052003 | www.linkedin.com/in/manavshah-11ai/

AI Engineer with AI Engineer with hands-on experience in LLMs, generative AI, and applied machine learning, building automation pipelines and AI-driven solutions for enterprise clients. Strong foundation in research, model evaluation, and end-to-end AI product development.

EDUCATIONAL QUALIFICATION

M.Tech in Artificial Intelligence — Pandit Deendayal Energy University CGPA: 9.06 | 2024–2026

B.Tech in Computer Engineering — Indus University CGPA: 9.70 | 2021–2024

WORK EXPERIENCE

Analytix Solutions | Ahmedabad / ***Associate AI Developer*** May 2026 – Present

- Owned TaxProGenie: automated 28+ U.S. tax forms across 1–1.5L taxpayers using LLM classification, Azure Document Intelligence, OpenAI/Claude extraction, validation & Excel automation; achieved 98% standard / 90% non-standard accuracy.
- Reduced 5–10 min/form → 1–2 min/PDF through batch, near-zero-intervention processing, automating a ~20-person tax-season workflow and eliminating repetitive manual entry.
- Built AI-VOX: Gemini TTS-based voice-training platform tested by 20 HR/L&D employees, generating 10-question sessions and evaluating pronunciation, fluency, vocabulary, confidence & sentiment at ~95% accuracy.
- Execute AI model R&D, prompt engineering, API integration, fine-tuning, benchmarking, evaluation, code review & deployment for client-facing AI solutions.

Analytix Solutions | Ahmedabad / ***AI Engineer Intern*** Nov 2025 – May 2026

- Built Artifax, an AI meeting-to-documentation pipeline that converts video meetings/demos into timestamped Word documentation, extracting audio, key workflow steps and contextual screenshots.
- Automated step-by-step workflow reconstruction + flow-diagram generation, linking each action to its exact timestamp and corresponding video frame for faster client handoffs and BRD/process documentation.

Schbang | Remote / ***AI Development Intern*** Jun 2025 – Aug 2025

- Developed AI agents for automated video analysis/categorization, multimedia web-scraping pipelines, and cross-modal image↔video transformation workflows for AI-driven creative applications.

TECHNICAL SKILLS

Languages: Python, SQL, C++
AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI
GenAI: LLMs, RAG, Prompt Engineering, LLM Evaluation, Vector Databases, AI Agents
Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, Streamlit, FastAPI
Cloud & APIs: Azure, AWS, OpenAI, Anthropic, Gemini, REST APIs, Git, GitHub

RESEARCH & PUBLICATIONS

EEGNet-BiLSTM with Gated Fusion for Sleep Stage Classification — *IEEE | 88.02% Accuracy*

- Performed data preparation, training, fine-tuning, hyperparameter optimization and benchmarking.
- Designed EEGNet-BiLSTM with gated fusion for cross-subject and cross-age sleep-stage classification.

DeepBoost — Multi-Stage Deep Feature Extraction for Sleep Apnea Detection — *IEEE | 90% Accuracy*

- Developed LSTM/RNN + XGBoost decision fusion for automated sleep-apnea detection.
- Engineered deep-feature extraction, model training, tuning and comparative benchmarking end-to-end.

PERSONAL / ACADEMIC PROJECTS

Applied AI OS — *Independent AI Engineering Workspace | 28+ AI-Powered Features | LLMs • RAG • Prompt Engineering • AI Evaluation*

- Built a Multi-LLM Playground + Prompt Studio supporting OpenAI, Anthropic & Gemini, with prompt versioning, A/B testing, RAG, response comparison and reusable templates.
- Engineered AI reliability & observability covering hallucination detection, JSON repair, experiments, token costs, API latency/health and developer tooling.
- Built reusable AI engineering utilities including POC Builder, RAG Research Assistant, Code Review, API testing, encrypted API Key Vault and automated documentation.

AI-Driven Smart Greenhouse for Automated Mushroom Cultivation — *ICCNT 2024 | ~87% Accuracy | Raspberry Pi 5*

- Trained YOLOv8 + ViT on 1,000–2,000 images, manually annotating mushroom data and achieving ~87% accuracy.
- Deployed the vision pipeline on Raspberry Pi 5 + Pi Camera with environmental-sensor integration for automated mushroom monitoring.

LEADERSHIP & TEACHING

Research Team Lead, M.Tech — Guided junior researchers on paper development, research methodology and organization.
Teaching Assistant, PDEU — Taught B.Tech students and resolved doubts in complex technical subjects.

Portfolio -manav-aiportfolio.netlify.app